# Xiaoqi Lin

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## Education

**Columbia University** New York, NY
*M.S. in Electrical Engineering* Dec. 2022

**Beijing Jiaotong University (BJTU)** Beijing, China
*B.S. in Electrical Engineering* Jun. 2021

## Research Experience

**Staff Associate (Computational Biology),** Shen Lab May 2023 – Present
Department of Systems Biology, Columbia University Irving Medical Center
**Advisors:** Yufeng Shen and Wendy Chung

- **Genetic architecture of human congenital anomalies**
  Investigated the genetic architecture of esophageal atresia with or without tracheoesophageal fistula (EA/TEF), a rare congenital foregut anomaly, using whole-genome and exome sequencing of 401 affected individuals.
  - Addressed the limited explanatory power of known risk genes and poorly characterized rare inherited variants by jointly analyzing *de novo* and rare inherited variants. Demonstrated that both protein-truncating and missense variants contribute to EA/TEF risk, with effect sizes positively correlated with selection strength.
  - Identified significant enrichment of *de novo* coding variants in metabolic processes, implicating new biological processes in human foregut development.
  - Proposed five novel candidate EA/TEF risk genes (FDR < 0.15) through rare variant association analysis against 47,419 population controls, including *APC2* (FDR < 0.05). These associations were mostly driven by rare inherited variants, consistent with a complex architecture in which many genes contribute large relative risk with moderate penetrance.
- **Deep learning models of noncoding variant effect on post-transcriptional regulation** *(ongoing)*
  - Estimating the fitness impact of noncoding variants from the human allele frequency spectrum, using RNA-binding protein (RBP) binding and cross-species conservation as priors; developing deep learning models that learn RBP binding across proteins to predict variant effects on post-transcriptional regulation.
  - Training sequence models of RBP binding on ENCODE eCLIP data (168 RBPs) and benchmarking variant effect scores against human allele frequency, and existing predictors (e.g. GPN-Star, AlphaGenome).
  - Applying the model to *de novo* variants in UTRs in autism and congenital heart disease cohorts to test for case enrichment and to relate predicted site disruption to selection coefficients outside conserved regions.
- **Applying Population Genetic Concepts to Genomic Medicine**
  Returning pathogenic variants in early-onset neurodevelopmental disorders often depends on confirmed *de novo* status, which lowers diagnostic yield in families without parental genotypes. Proposed selection coefficients (*S*) as an inheritance-independent criterion to prioritize pathogenic variants.
  - Quantified selection coefficients for 847 clinically returned pathogenic/likely pathogenic variants in 21,532 individuals from the SPARK autism cohort, showing that pathogenic variants have substantially higher *S* than ultra-rare background variants. 98% of loss-of-function and 67% of missense variants ranked in the top three within each individual.
  - Demonstrated that *de novo* variants have higher *S* than rare inherited. Stringent *S* (MisFit) thresholds recovered *de novo* enrichment in non-trio cases, including sex and cognition status specific thresholds that match or exceed enrichment observed for *de novo* variants.
  - Evaluated *S*-based ranking across autism (SPARK) and congenital heart disease (PCGC) cohorts, showing that *S* captures cohort specific, biologically meaningful signal.
- **Phasing Pathogenic Variants in KIF1A to Facilitate Gene Therapy**
  - Applied a scalable strategy to selectively knock down *de novo* missense *KIF1A* while sparing wild-type, enabling a polymorphism-targeted antisense oligonucleotide (ASO) strategy instead of bespoke, variant-specific therapies.

- Developed an automated workflow using long-read sequencing data to infer *cis* relationships between pathogenic *KIF1A* variants and nearby common SNPs, and performed an exhaustive cross-patient search that identified four shared SNPs which, if targetable, would allow 54 individuals to receive allele-specific ASO therapy.
- Performed coverage modeling showing that targeting these shared SNP sites could enable allele-specific intervention for ∼89% of patients, with a 1000 Genomes–based greedy analysis indicating that targeting 3 SNP sites could reach ≈80% of individuals globally and 6 sites >90%, supporting the feasibility of a limited ASO set targeting shared polymorphisms.

- **Data Curation and Bioinformatics Pipelines for Early-onset Disease Analysis**
  - Maintained and extended Mutable.wiki, a public, gene-centric database featuring 79,000+ curated *de novo* variants from published studies of neurodevelopmental disorders and congenital anomalies; implemented gene-based querying, variant-level annotation, and 3D visualization of spatial clustering among missense variants.
  - Optimized variant analysis pipelines for both database curation and clinical diagnostics, supporting multiple input formats (FASTQ, BAM/CRAM, VCF). Enhancements include improved variant calling, quality control, and functional annotation using gene- and variant-level constraint scores to support identification of rare disease variants in early-onset cases.

**Research Engineer Intern,** Teradata Corporation May 2022 – Aug. 2022
**Advisor:** Christopher Antoun

- Processed database query logs into graph objects, leveraging StellarGraph to encode categorical variables and scale values for accurate analysis. Applied graph embeddings and BIRCH clustering to identify and visualize patterns in customer query behavior.
- Developed and implemented a machine learning classifier to predict query execution plan (QEP) performance, enabling the selection of optimized plans to prevent potential performance regressions with a loss rate below 0.5%.

**Research Assistant,** Internet Real-Time Lab Dec. 2021 – May 2022
Department of Computer Science, Columbia University
**Advisors:** Henning Schulzrinne and Gaston Ormazabal

- Processed network vectors with Lasso regression, reducing redundant attributes from 107 to 14 features.
- Utilized Random Forest and K-Neighbors classifiers to categorize device behaviors and detected data collection patterns with accuracy above 95%.

# Publications and Preprints

## Published

Michael V. Zuccaro, Randee E. Young, Jiangyuan Hu, Patricia Lanzano, Ekaterina Semenova, **Xiaoqi Lin**, Alicja Skowronski, Yufeng Shen, Tae Hyun Kim, Noelle Germain, Patrick Sarmiere, Charles A. LeDuc, Danny E. Miller, Wendy K. Chung[*]. *Antisense oligonucleotides to KIF1A polymorphisms expand targets and rescue patient-derived neurons in vitro.* **Nature Communications** 17, 1109 (2026).

Yuan Zhang, Mian Umair Ahsan, Peng Wang, **Xiaoqi Lin**, Ian M. Campbell, Cong Liu, Wendy K. Chung, Chunhua Weng, Kai Wang[*]. *RankVar: Machine learning-based variant ranking and reinterpretation for rare genetic diseases.* **Genome Medicine** 18, 139 (2026). doi:10.1186/s13073-026-01701-2

## Manuscripts under review

Lu Qiao[†], Pamela Feliciano[†], Jacob Hagen, Chao Hou, **Xiaoqi Lin**, Abigail Wang, The SPARK Consortium, Yufeng Shen[*], Wendy K. Chung[*]. *Rare inherited variants explain a third of attributable risk for autism.* In revision at **Nature**.

**Xiaoqi Lin**[†], Shangqian Xie[†], Elaine Cui, Rebecca Hernan, Priyanka Ahimaz, Jacob J. Hagen, Christophe Faure, Lu Qiao, Paul Kingma, Nicole A. Edwards, William Middlesworth, Julie Khlevner, Mahmoud El Fiky, David Schindel, Elizabeth Fialkowski, Aaron M. Zorn, Yufeng Shen[*], Wendy K. Chung[*]. *Rare variants highlight new genetic contributors and metabolic pathways in esophageal atresia and tracheoesophageal fistula.* Submitted and under peer-review

### Manuscripts in Preparation

**Xiaoqi Lin**, Jessica R. Wright, Lu Qiao, Wenxing Li, Jacob Hagen, Pamela Feliciano, Wendy K. Chung, Yufeng Shen. *Using selection coefficients to classify novel missense variants as likely pathogenic in neurodevelopmental conditions.*

Zining Chen[†], Jacob Hagen[†], **Xiaoqi Lin**, Wendy K. Chung, Yufeng Shen. *Mutable: An integrated web application for visualization and analysis of de novo variants in human diseases.*

[†]Equal contribution. [*]Corresponding author.

## Presentations

- **Poster Presentation**, *American Society of Human Genetics Annual Meeting 2025*
  *Using selection coefficients to classify novel missense variants as likely pathogenic in neurodevelopmental conditions.*
- **Poster Presentation**, *American Society of Human Genetics Annual Meeting 2024*
  *Estimated selection coefficients of protein coding variants support large fitness impact of pathogenic variants in neurodevelopmental disorders.*

## Teaching Experience

**Mentor,** Summer Student Research Program — Summers 2025, 2026
Department of Systems Biology, Columbia University
Supervised five summer students across two summers through research projects, focusing on project design, coding workflows, and presentation skills.

**Teaching Assistant,** Computer Networks — Sep. 2022 – May 2023
Department of Computer Science, Columbia University
**Advisor:** Vishal Misra
Provided one-on-one support through office hours on coursework and exam topics; responsible for grading assignments and assisting with course logistics.

**Teaching Assistant,** Electrical Circuits — Sep. 2019 – Dec. 2019
Beijing Jiaotong University
**Advisor:** Yin Xu
Assisted with lab experiments and held office hours to support students with lab reports.

## Peer Review

**PLoS Computational Biology** — Invited Reviewer, 2024

## Skills

**Genomic and genetic data analysis:** Nextflow, SAMtools, BCFtools, FreeBayes, DeepVariant
**Programming:** Python, Bash, SQL, HTML, JavaScript
**Frameworks & Tools:** PyTorch, TensorFlow, Keras, sklearn, Docker, AWS, Kubernetes, MongoDB, Databricks